Chapter 6: Circular Motion and Other Applications of Newton's Laws

6.1 Extending the Particle in Uniform Circular Motion Model

6.2 Nonuniform Circular Motion

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6.1 Extending the Particle in Uniform Circular Motion Model

In Section 4.4, we discussed the analysis model of a particle in uniform circular motion, in which a particle moves with constant speed v in a circular path having a radius r. The particle experiences an acceleration that has a magnitude

$$a_c = \frac{v^2}{r}$$

The acceleration is called *centripetal acceleration* because $\vec{\mathbf{a}}_c$ is directed toward the center of the circle. Furthermore, $\vec{\mathbf{a}}_c$ is *always* perpendicular to $\vec{\mathbf{v}}$. (If there were a component of acceleration parallel to $\vec{\mathbf{v}}$, the particle's speed would be changing.)

Let us now extend the particle in uniform circular motion model from Section 4.4 by incorporating the concept of force. Consider a puck of mass m that is tied to a string of length r and moves at constant speed in a horizontal, circular path as illustrated in Figure 6.1.

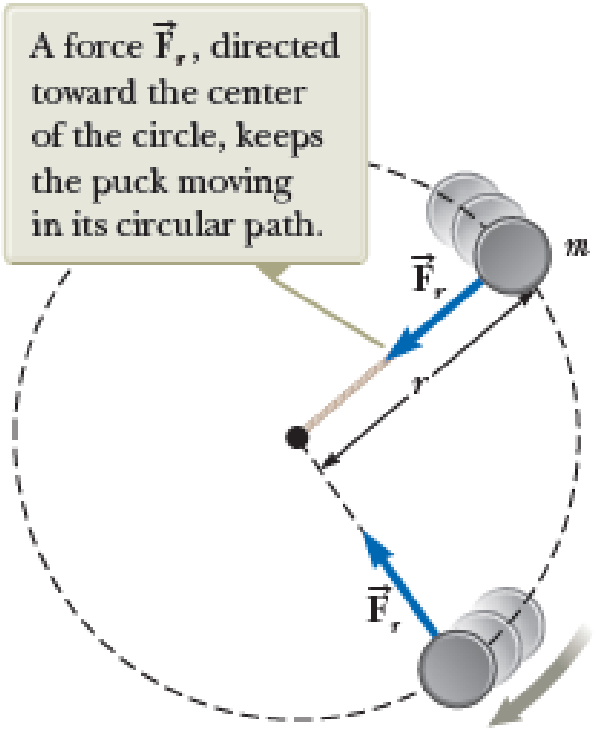


Figure 6.1 An overhead view of a puck moving in a circular path in a horizontal plane.

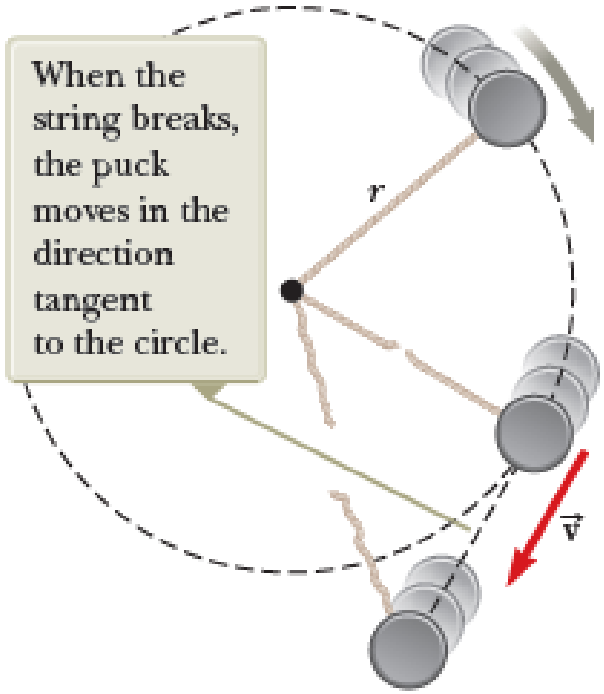


Figure 6.2 The string holding the puck in its circular path breaks.

If Newton's second law is applied along the radial direction, the net force causing the centripetal acceleration can be related to the acceleration as follows:

$$\sum F = ma_c = m\frac{v^2}{r} \qquad (6.1)$$

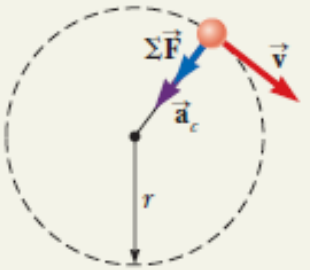
A force causing a centripetal acceleration acts toward the center of the circular path and causes a change in the direction of the velocity vector. If that force should vanish, the object would no longer move in its circular path; instead, it would move along a straight-line path tangent to the circle. This idea is illustrated in Figure 6.2 for the puck moving in a circular path at the end of a string in a horizontal plane. If the string breaks at some instant, the puck moves along the straight-line path that is tangent to the circle at the position of the puck at this instant.

Analysis Model **Particle in Uniform Circular Motion (Extension)**

Imagine a moving object that can be modeled as a particle. If it moves in a circular path of radius r at a constant speed v, it experiences a centripetal acceleration. Because the particle is accelerating, there must be a net force acting on the particle. That force is directed toward the center of the circular path and is given by

$$\sum F = ma_c = m\frac{v^2}{r} \qquad (6.1)$$

Examples



- the tension in a string of constant length acting on a rock twirled in a circle
- the gravitational force acting on a planet traveling around the Sun in a perfectly circular orbit (Chapter 13)
- the magnetic force acting on a charged particle moving in a uniform magnetic field (Chapter 29)
- the electric force acting on an electron in orbit around a nucleus in the Bohr model of the hydrogen atom (Chapter 42)

Example 6.1 The Conical Pendulum AM

A small ball of mass m is suspended from a string of length L. The ball revolves with constant speed v in a horizontal circle of radius r as shown in Figure 6.3. (Because the string sweeps out the surface of a cone, the system is known as a *conical pendulum.*) Find an expression for v in terms of the geometry in Figure 6.3.

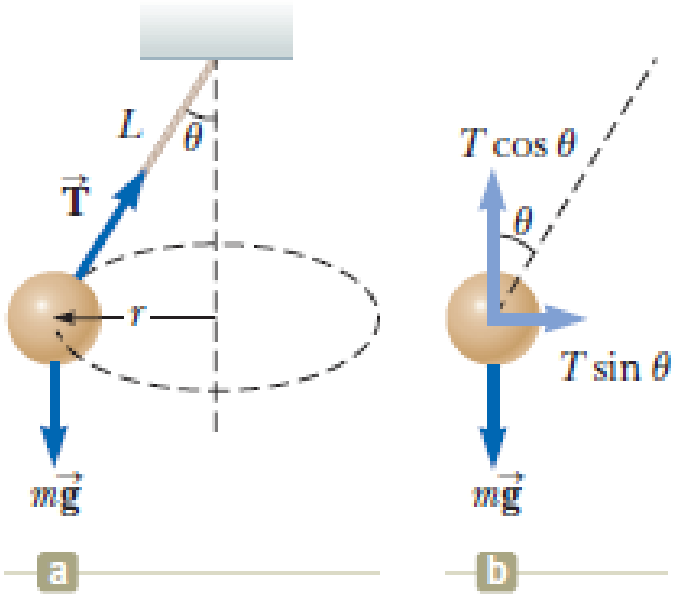


Figure 6.3 (Example 6.1) (a) A conical pendulum. The path of the ball is a horizontal circle. (b) The forces acting on the ball.

$$\sum F_y = T\cos\theta - mg = 0$$

(1) $T\cos\theta = mg$

(2) $\sum F_x = T\sin\theta = ma_c = \dfrac{mv^2}{r}$

$$\tan\theta = \frac{v^2}{rg}$$

$r = L\sin\theta$

$$v = \sqrt{rg\tan\theta}$$

$$v = \sqrt{Lg\sin\theta\tan\theta}$$

Example 6.2 How Fast Can It Spin? AM

A puck of mass 0.500 kg is attached to the end of a cord 1.50 m long. The puck moves in a horizontal circle as shown in Figure 6.1. If the cord can withstand a maximum tension of 50.0 N, what is the maximum speed at which the puck can move before the cord breaks? Assume the string remains horizontal during the motion.

$$T = m\frac{v^2}{r}$$

$$(1) \quad v = \sqrt{\frac{Tr}{m}}$$

$$v_{\max} = \sqrt{\frac{T_{\max} r}{m}} = \sqrt{\frac{(50.0\ \text{N})(1.50\ \text{m})}{0.500\ \text{kg}}} = 12.2\ \text{m/s}$$

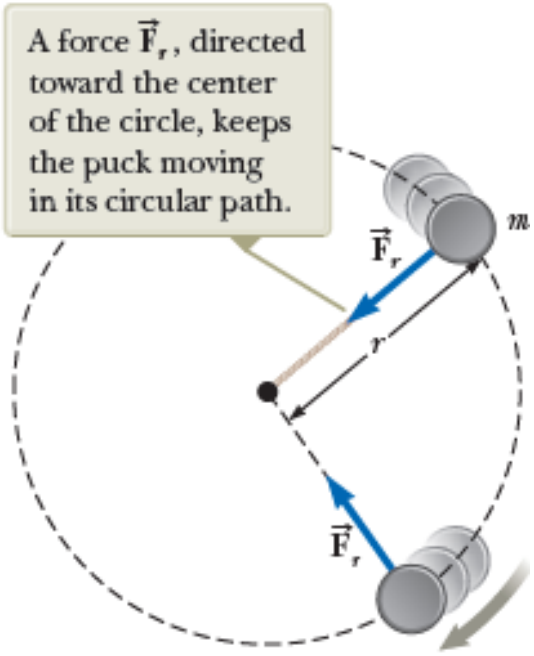


Figure 6.1 An overhead view of a puck moving in a circular path in a horizontal plane.

Finalize Equation (1) shows that v increases with T and decreases with larger m, as we expected from our conceptualization of the problem.

Example 6.3 What Is the Maximum Speed of the Car?

AM

A 1 500-kg car moving on a flat, horizontal road negotiates a curve as shown in Figure 6.4a. If the radius of the curve is 35.0 m and the coefficient of static friction between the tires and dry pavement is 0.523, find the maximum speed the car can have and still make the turn successfully.



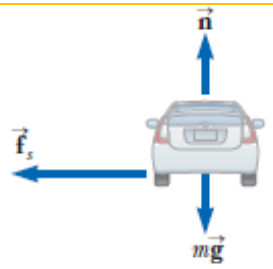


Figure 6.4 (Example 6.3) (a) The force of static friction directed toward the center of the curve keeps the car moving in a circular path. (b) The forces acting on the car.

$$(1) \quad f_{s,\max} = \mu_s n = m\frac{v_{\max}^2}{r}$$

$$\sum F_y = 0 \rightarrow n - mg = 0 \rightarrow n = mg$$

$$(2) \quad v_{\max} = \sqrt{\frac{\mu_s n r}{m}} = \sqrt{\frac{\mu_s m g r}{m}} = \sqrt{\mu_s g r}$$

$$v_{\max} = \sqrt{(0.523)(9.80\ \text{m/s}^2)(35.0\ \text{m})} = 13.4\ \text{m/s}$$

WHAT IF? Suppose a car travels this curve on a wet day and begins to skid on the curve when its speed reaches only 8.00 m/s. What can we say about the coefficient of static friction in this case?

Answer The coefficient of static friction between the tires and a wet road should be smaller than that between the tires and a dry road. This expectation is consistent with experience with driving because a skid is more likely on a wet road than a dry road.

To check our suspicion, we can solve Equation (2) for the coefficient of static friction:

$$\mu_s = \frac{v_{\max}^2}{gr}$$

Substituting the numerical values gives

$$\mu_s = \frac{v_{\max}^2}{gr} = \frac{(8.00\ \text{m/s})^2}{(9.80\ \text{m/s}^2)(35.0\ \text{m})} = 0.187$$

which is indeed smaller than the coefficient of 0.523 for the dry road.

Example 6.4 The Banked Roadway

AM

A civil engineer wishes to redesign the curved roadway in Example 6.3 in such a way that a car will not have to rely on friction to round the curve without skidding. In other words, a car moving at the designated speed can negotiate the curve even when the road is covered with ice. Such a road is usually *banked*, which means that the roadway is tilted toward the inside of the curve as seen in the opening photograph for this chapter. Suppose the designated speed for the road is to be 13.4 m/s (30.0 mi/h) and the radius of the curve is 35.0 m. At what angle should the curve be banked?

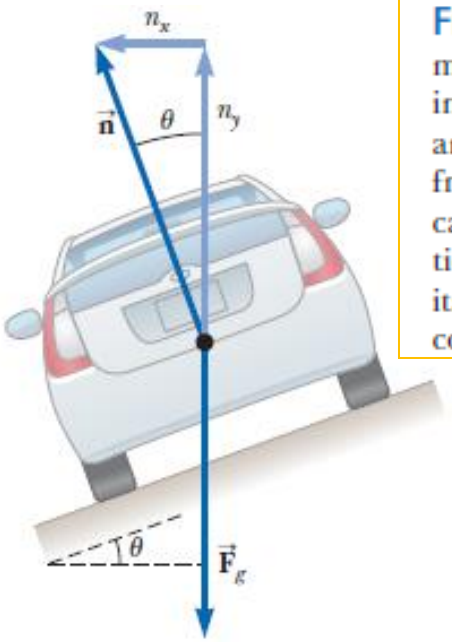


Figure 6.5 (Example 6.4) A car moves into the page and is rounding a curve on a road banked at an angle θ to the horizontal. When friction is neglected, the force that causes the centripetal acceleration and keeps the car moving in its circular path is the horizontal component of the normal force.

$$(1)\quad \sum F_r = n\sin\theta = \frac{mv^2}{r}$$

$$\sum F_y = n\cos\theta - mg = 0$$

$$(2)\quad n\cos\theta = mg$$

$$(3)\quad \tan\theta = \frac{v^2}{rg}$$

$$\theta = \tan^{-1}\left[\frac{(13.4\ \text{m/s})^2}{(35.0\ \text{m})(9.80\ \text{m/s}^2)}\right] = 27.6^\circ$$

Example 6.5 Riding the Ferris Wheel

AM

A child of mass m rides on a Ferris wheel as shown in Figure 6.6a. The child moves in a vertical circle of radius 10.0 m at a constant speed of 3.00 m/s.

(A) Determine the force exerted by the seat on the child at the bottom of the ride. Express your answer in terms of the weight of the child, mg.

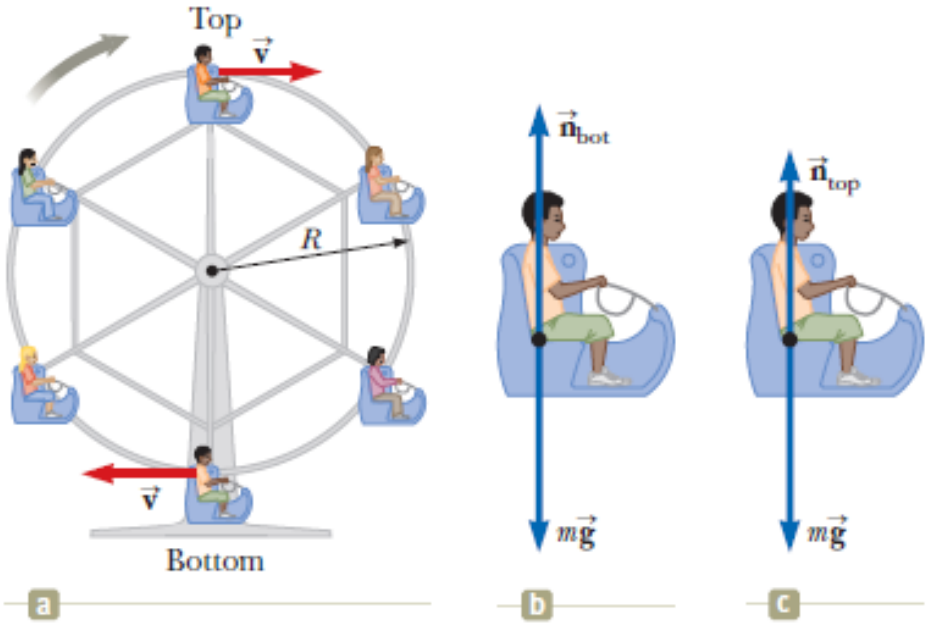


Figure 6.6 (Example 6.5) (a) A child rides on a Ferris wheel. (b) The forces acting on the child at the bottom of the path. (c) The forces acting on the child at the top of the path.

$$\sum F = n_{bot} - mg = m\frac{v^2}{r}$$

$$n_{bot} = mg + m\frac{v^2}{r} = mg\left(1 + \frac{v^2}{rg}\right)$$

$$n_{bot} = mg\left[1 + \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$

$$= 1.09\,mg$$

Hence, the magnitude of the force $\vec{n}_{bot}$ exerted by the seat on the child is *greater* than the weight of the child by a factor of 1.09. So, the child experiences an apparent weight that is greater than his true weight by a factor of 1.09.

(B) Determine the force exerted by the seat on the child at the top of the ride.

$$\sum F = mg - n_{top} = m\frac{v^2}{r}$$

$$n_{top} = mg - m\frac{v^2}{r} = mg\left(1 - \frac{v^2}{rg}\right)$$

$$n_{top} = mg\left[1 - \frac{(3.00\ \text{m/s})^2}{(10.0\ \text{m})(9.80\ \text{m/s}^2)}\right]$$

$$= 0.908\,mg$$

In this case, the magnitude of the force exerted by the seat on the child is *less* than his true weight by a factor of 0.908, and the child feels lighter.

6.2 Nonuniform Circular Motion

In Chapter 4, we found that if a particle moves with varying speed in a circular path, there is, in addition to the radial component of acceleration, a tangential component having magnitude $|dv/dt|$. Therefore, the force acting on the particle must also have a tangential and a radial component. Because the total acceleration is $\vec{\mathbf{a}} = \vec{\mathbf{a}}_r + \vec{\mathbf{a}}_t$, the total force exerted on the particle is $\sum \vec{\mathbf{F}} = \sum \vec{\mathbf{F}}_r + \sum \vec{\mathbf{F}}_t$ as shown in Figure 6.7. (We express the radial and tangential forces as net forces with the summation notation because each force could consist of multiple forces that combine.) The vector $\sum \vec{\mathbf{F}}_r$ is directed toward the center of the circle and is responsible for the centripetal acceleration. The vector $\sum \vec{\mathbf{F}}_t$ tangent to the circle is responsible for the tangential acceleration, which represents a change in the particle's speed with time.

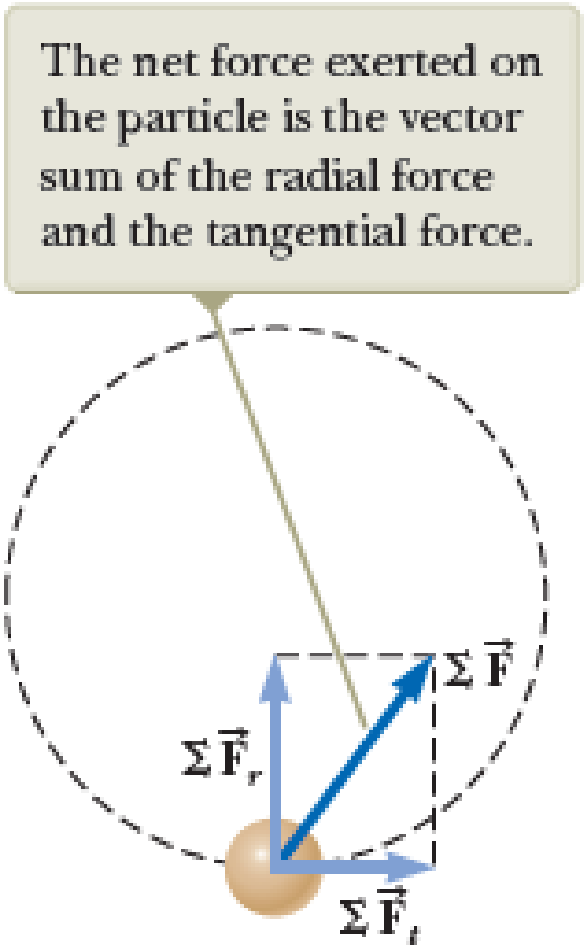


Figure 6.7 When the net force acting on a particle moving in a circular path has a tangential component ΣF_t, the particle's speed changes.

Example 6.6 Keep Your Eye on the Ball

AM

A small sphere of mass m is attached to the end of a cord of length R and set into motion in a *vertical* circle about a fixed point O as illustrated in Figure 6.9. Determine the tangential acceleration of the sphere and the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.

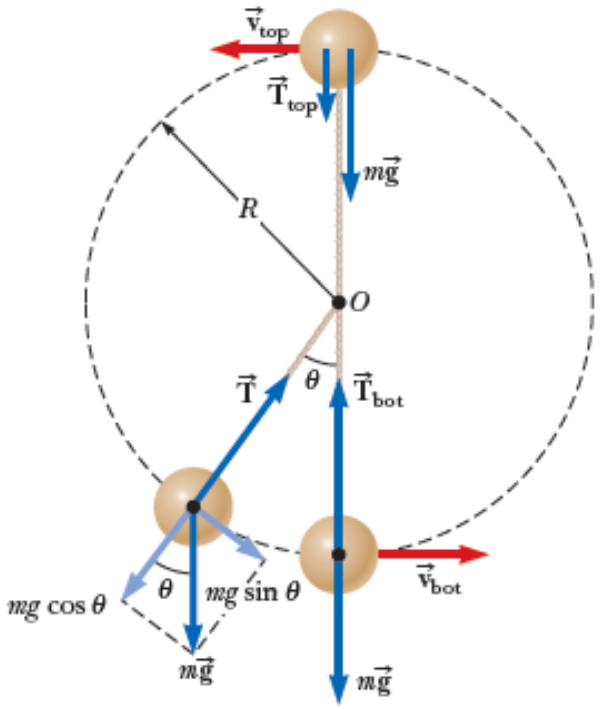


Figure 6.9 (Example 6.6) The forces acting on a sphere of mass m connected to a cord of length R and rotating in a vertical circle centered at O. Forces acting on the sphere are shown when the sphere is at the top and bottom of the circle and at an arbitrary location.

$$\sum F_t = mg\sin\theta = ma_t$$

$$a_t = g\sin\theta$$

$$\sum F_r = T - mg\cos\theta = \frac{mv^2}{R}$$

$$T = mg\left(\frac{v^2}{Rg} + \cos\theta\right)$$

$$T_{\text{top}} = mg\left(\frac{v_{\text{top}}^2}{Rg} - 1\right)$$

$$T_{\text{bot}} = mg\left(\frac{v_{\text{bot}}^2}{Rg} + 1\right)$$

WHAT IF? What if the ball is set in motion with a slower speed?

(A) What speed would the ball have as it passes over the top of the circle if the tension in the cord goes to zero instantaneously at this point?

Answer Let us set the tension equal to zero in the expression for T_{top}:

$$0 = mg\left(\frac{v_{\text{top}}^2}{Rg} - 1\right) \rightarrow v_{\text{top}} = \sqrt{gR}$$

(B) What if the ball is set in motion such that the speed at the top is less than this value? What happens?

Answer In this case, the ball never reaches the top of the circle. At some point on the way up, the tension in the string goes to zero and the ball becomes a projectile. It follows a segment of a parabolic path over the top of its motion, rejoining the circular path on the other side when the tension becomes nonzero again.

Problems with Solutions

1. AMT M A light string can support a stationary hanging load of 25.0 kg before breaking. An object of mass $m = 3.00$ kg attached to the string rotates on a frictionless, horizontal table in a circle of radius $r = 0.800$ m, and the other end of the string is held fixed as in Figure P6.1. What range of speeds can the object have before the string breaks?

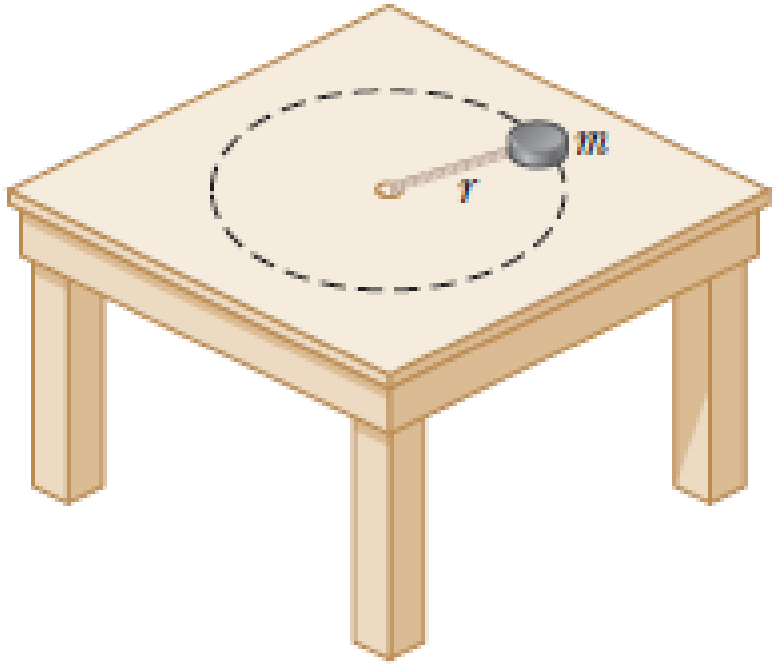


Figure P6.1

We are given $m = 3.00$ kg, $r = 0.800$ m. The string will break if the tension exceeds the weight corresponding to 25.0 kg, so

$$T_{max} = Mg = (25.0 \text{ kg})(9.80 \text{ m/s}^2) = 245 \text{ N}$$

When the 3.00-kg mass rotates in a horizontal circle, the tension causes the centripetal acceleration,

so $$T = \frac{mv^2}{r}$$

Then

$$v^2 = \frac{rT}{m} = \frac{(0.800 \text{ m})T}{3.00 \text{ kg}} \leq \frac{(0.800 \text{ m})T_{max}}{3.00 \text{ kg}}$$

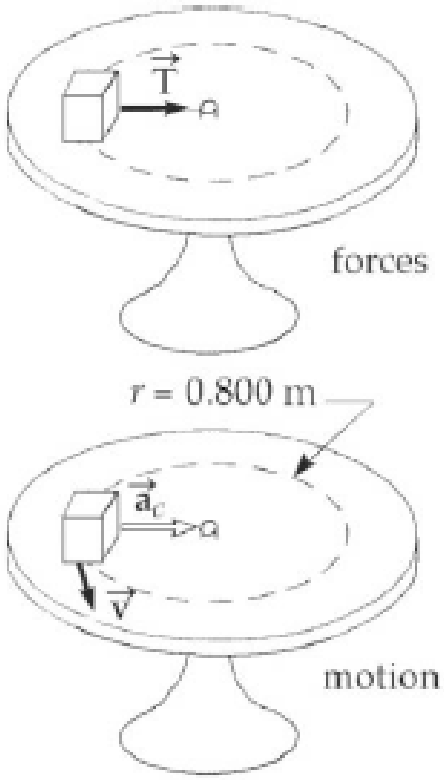
$$= \frac{(0.800 \text{ m})(245 \text{ N})}{3.00 \text{ kg}} = 65.3 \text{ m}^2/\text{s}^2$$

This represents the maximum value of v^2, or

$$0 \leq v \leq \sqrt{65.3} \text{ m/s}$$

which gives

$$\boxed{0 \leq v \leq 8.08 \text{ m/s}}$$



ANS. FIG. P6.1

2. Whenever two *Apollo* astronauts were on the surface of the Moon, a third astronaut orbited the Moon. Assume the orbit to be circular and 100 km above the surface of the Moon, where the acceleration due to gravity is 1.52 m/s^2. The radius of the Moon is 1.70×10^6 m. Determine (a) the astronaut's orbital speed and (b) the period of the orbit.

(a) The astronaut's orbital speed is found from Newton's second law, with

$$\sum F_y = ma_y: \; mg_{moon} \text{ down} = \frac{mv^2}{r} \text{ down}$$

solving for the velocity gives

$$v = \sqrt{g_{moon} r} = \sqrt{\left(1.52 \text{ m/s}^2\right)\left(1.7 \times 10^6 \text{ m} + 100 \times 10^3 \text{ m}\right)}$$

$$v = \boxed{1.65 \times 10^3 \text{ m/s}}$$

(b) To find the period, we use $v = \dfrac{2\pi r}{T}$ and solve for T:

$$T = \frac{2\pi\left(1.8 \times 10^6 \text{ m}\right)}{1.65 \times 10^3 \text{ m/s}} = \boxed{6.84 \times 10^3 \text{ s}} = 1.90 \text{ h}$$

3. In the Bohr model of the hydrogen atom, an electron moves in a circular path around a proton. The speed of the electron is approximately 2.20×10^6 m/s. Find (a) the force acting on the electron as it revolves in a circular orbit of radius 0.529×10^{-10} m and (b) the centripetal acceleration of the electron.

(a) The force acting on the electron in the Bohr model of the hydrogen atom is directed radially inward and is equal to

$$F = \frac{mv^2}{r} = \frac{(9.11\times10^{-31}\ \text{kg})(2.20\times10^{6}\ \text{m/s})^2}{0.529\times10^{-10}\ \text{m}}$$

$$= \boxed{8.33\times10^{-8}\ \text{N inward}}$$

(b) $$a = \frac{v^2}{r} = \frac{(2.20\times10^{6}\ \text{m/s})^2}{0.529\times10^{-10}\ \text{m}} = \boxed{9.15\times10^{22}\ \text{m/s}^2\ \text{inward}}$$

4. A curve in a road forms part of a horizontal circle. As a car goes around it at constant speed 14.0 m/s, the total horizontal force on the driver has magnitude 130 N.

What is the total horizontal force on the driver if the speed on the same curve is 18.0 m/s instead?

In $\sum F = m\frac{v^2}{r}$, both m and r are unknown but remain constant.

Symbolically, write

$$\sum F_{slow} = \left(\frac{m}{r}\right)(14.0\ \text{m/s})^2 \text{ and } \sum F_{fast} = \left(\frac{m}{r}\right)(18.0\ \text{m/s})^2$$

Therefore, $\sum F$ is proportional to v^2 and increases by a factor of $\left(\frac{18.0}{14.0}\right)^2$ as v increases from 14.0 m/s to 18.0 m/s. The total force at the higher speed is then

$$\sum F_{fast} = \left(\frac{18.0}{14.0}\right)^2 \sum F_{slow} = \left(\frac{18.0}{14.0}\right)^2 (130\ \text{N}) = \boxed{215\ \text{N}}$$

This force must be **horizontally inward** to produce the driver's centripetal acceleration.

5. In a cyclotron (one type of particle accelerator), a deuteron (of mass 2.00 u) reaches a final speed of 10.0% of the speed of light while moving in a circular path of radius 0.480 m. What magnitude of magnetic force is required to maintain the deuteron in a circular path?

We neglecting relativistic effects. With $1\ u = 1.661 \times 10^{-27}$ kg, and from Newton's second law, we obtain

$$F = ma_c = \frac{mv^2}{r}$$

$$= \left(2 \times 1.661 \times 10^{-27}\ \text{kg}\right) \frac{\left(2.998 \times 10^{7}\ \text{m/s}\right)^2}{(0.480\ \text{m})}$$

$$= \boxed{6.22 \times 10^{-12}\ \text{N}}$$

6. A car initially traveling **W** eastward turns north by traveling in a circular path at uniform speed as shown in Figure P6.6. The length of the arc *ABC* is 235 m, and the car completes the turn in 36.0 s. (a) What is the acceleration when the car is at *B* located at an angle of 35.0°? Express your answer in terms of the unit vectors $\hat{i}$ and $\hat{j}$. Determine (b) the car's average speed and (c) its average acceleration during the 36.0-s interval.

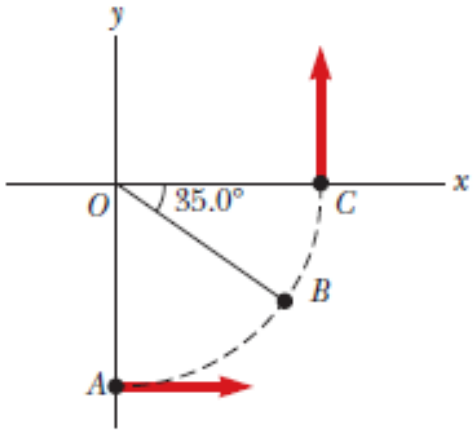


Figure P6.6

(a) The car's speed around the curve is found from

$$v = \frac{235\ \text{m}}{36.0\ \text{s}} = 6.53\ \text{m/s}$$

This is the answer to part (b) of this problem. We calculate the radius of the curve from $\frac{1}{4}(2\pi r) = 235\ \text{m}$, which gives $r = 150\ \text{m}$.

The car's acceleration at point B is then

$$\vec{a}_r = \left(\frac{v^2}{r}\right) \text{ toward the center}$$

$$= \frac{(6.53 \text{ m/s})^2}{150 \text{ m}} \text{ at } 35.0^\circ \text{ north of west}$$

$$= (0.285 \text{ m/s}^2)\left(\cos 35.0^\circ(-\hat{i}) + \sin 35.0^\circ \hat{j}\right)$$

$$= \boxed{\left(-0.233\hat{i} + 0.163\hat{j}\right) \text{ m/s}^2}$$

(b) From part (a), $v = \boxed{6.53 \text{ m/s}}$

(c) We find the average acceleration from

$$\vec{a}_{avg} = \frac{\left(\vec{v}_f - \vec{v}_i\right)}{\Delta t}$$

$$= \frac{\left(6.53\hat{j} - 6.53\hat{i}\right) \text{ m/s}}{36.0 \text{ s}}$$

$$= \boxed{\left(-0.181\hat{i} + 0.181\hat{j}\right) \text{ m/s}^2}$$

7. A space station, in the form of a wheel 120 m in diameter, rotates to provide an "artificial gravity" of 3.00 m/s^2 for persons who walk around on the inner wall of the outer rim. Find the rate of the wheel's rotation in revolutions per minute that will produce this effect.

Standing on the inner surface of the rim, and moving with it, each person will feel a normal force exerted by the rim. This inward force causes the 3.00 m/s^2 centripetal acceleration:

$$a_c = v^2/r \quad \text{so} \quad v = \sqrt{a_c r} = \sqrt{(3.00\ \text{m/s}^2)(60.0\ \text{m})} = 13.4\ \text{m/s}$$

The period of rotation comes from $v = \frac{2\pi r}{T}$:

$$T = \frac{2\pi r}{v} = \frac{2\pi(60.0\ \text{m})}{13.4\ \text{m/s}} = 28.1\ \text{s}$$

so the frequency of rotation is

$$f = \frac{1}{T} = \frac{1}{28.1\ \text{s}} = \left(\frac{1}{28.1\ \text{s}}\right)\left(\frac{60\ \text{s}}{1\ \text{min}}\right) = \boxed{2.14\ \text{rev/min}}$$

8. Consider a conical pendulum (Fig. P6.8) with a bob **W** of mass $m = 80.0$ kg on a string of length $L = 10.0$ m that makes an angle of $\theta = 5.00°$ with the vertical. Determine (a) the horizontal and vertical components of the force exerted by the string on the pendulum and (b) the radial acceleration of the bob.

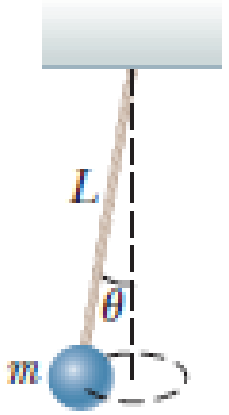


Figure P6.8

(a) The forces acting on the pendulum in the vertical direction must be in balance since the acceleration of the bob in this direction is zero. From Newton's second law in the y direction,

$$\sum F_y = T\cos\theta - mg = 0$$

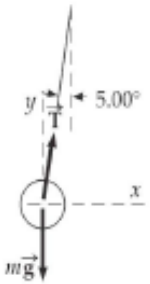
Solving for the tension T gives

$$T = \frac{mg}{\cos\theta} = \frac{(80.0\text{ kg})(9.80\text{ m/s}^2)}{\cos 5.00°} = 787\text{ N}$$

In vector form,

$$\vec{T} = T\sin\theta\,\hat{i} + T\cos\theta\,\hat{j}$$

$$= \boxed{(68.6\text{ N})\hat{i} + (784\text{ N})\hat{j}}$$



ANS. FIG. P6.8

(b) From Newton's second law in the x direction,

$$\sum F_x = T\sin\theta = ma_c$$

which gives

$$a_c = \frac{T\sin\theta}{m} = \frac{(787\text{ N})\sin 5.00°}{80.0\text{ kg}} = \boxed{0.857\text{ m/s}^2}$$

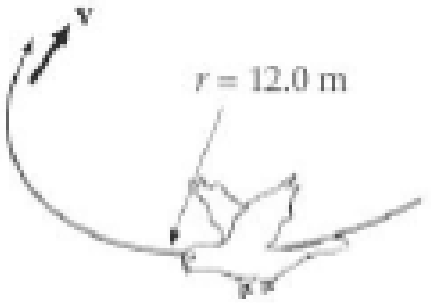
toward the center of the circle.

The length of the wire is unnecessary information. We could, on the other hand, use it to find the radius of the circle, the speed of the bob, and the period of the motion.

13. A hawk flies in a horizontal arc of radius 12.0 m at constant speed 4.00 m/s. (a) Find its centripetal acceleration. (b) It continues to fly along the same horizontal arc, but increases its speed at the rate of 1.20 m/s^2. Find the acceleration (magnitude and direction) in this situation at the moment the hawk's speed is 4.00 m/s.

(a) The hawk's centripetal acceleration is

$$a_c = \frac{v^2}{r} = \frac{(4.00\ \text{m/s})^2}{12.0\ \text{m}} = \boxed{1.33\ \text{m/s}^2}$$



ANS. FIG. P6.13

(b) The magnitude of the acceleration vector is

$$a = \sqrt{a_c^2 + a_t^2}$$

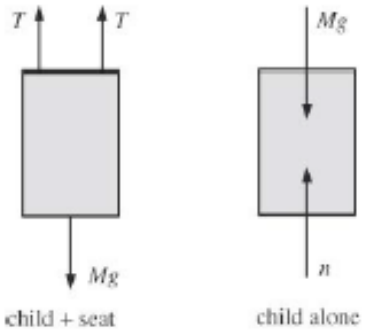
$$= \sqrt{(1.33\ \text{m/s}^2)^2 + (1.20\ \text{m/s}^2)^2} = \boxed{1.79\ \text{m/s}^2}$$

at an angle

$$\theta = \tan^{-1}\left(\frac{a_c}{a_t}\right) = \tan^{-1}\left(\frac{1.33\ \text{m/s}^2}{1.20\ \text{m/s}^2}\right) = \boxed{48.0^\circ \text{ inward}}$$

14. M A 40.0-kg child swings in a swing supported by two chains, each 3.00 m long. The tension in each chain at the lowest point is 350 N. Find (a) the child's speed at the lowest point and (b) the force exerted by the seat on the child at the lowest point. (Ignore the mass of the seat.)

We first draw a force diagram that shows the forces acting on the child-seat system and apply Newton's second law to solve the problem. The child's path is an arc of a circle, since the top ends of the chains are fixed. Then at the lowest point the child's motion is changing in direction: He moves with centripetal acceleration even as his speed is not changing and his tangential acceleration is zero.



ANS. FIG. P6.14

(a) ANS. FIG. P6.14 shows that the only forces acting on the system of child + seat are the tensions in the two chains and the weight of the boy:

$$\sum F = F_{net} = 2T - mg = ma = \frac{mv^2}{r}$$

with

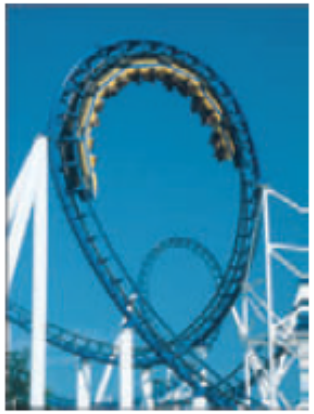
$$F_{net} = 2T - mg = 2(350\text{ N}) - (40.0\text{ kg})(9.80\text{ m/s}^2) = 308\text{ N}$$

solving for v gives

$$v = \sqrt{\frac{F_{net}r}{m}} = \sqrt{\frac{(308\text{ N})(3.00\text{ m})}{40.0\text{ kg}}} = \boxed{4.81\text{ m/s}}$$

(b) The normal force from the seat on the child accelerates the child in the same way that the total tension in the chain accelerates the child-seat system. Therefore, $n = 2T = \boxed{700\text{ N}}$.

17. A roller coaster at the Six Flags Great America amusement park in Gurnee, Illinois, incorporates some clever design technology and some basic physics. Each vertical loop, instead of being circular, is shaped like a teardrop (Fig. P6.17). The cars ride on the inside of the loop at the top, and the speeds are fast enough to ensure the cars remain on the track. The biggest loop is 40.0 m high. Suppose the speed at the top of the loop is 13.0 m/s and the corresponding centripetal acceleration of the riders is $2g$. (a) What is the radius of the arc of the teardrop at the top? (b) If the total mass of a car plus the riders is M, what force does the rail exert on the car at the top? (c) Suppose the roller coaster had a circular loop of radius 20.0 m. If the cars have the same speed, 13.0 m/s at the top, what is the centripetal acceleration of the riders at the top? (d) Comment on the normal force at the top in the situation described in part (c) and on the advantages of having teardrop-shaped loops.

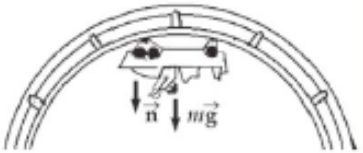


Frank Cezus/Getty Images

Figure P6.17

(a) $a_c = \dfrac{v^2}{r}$

$$r = \frac{v^2}{a_c} = \frac{(13.0\ \text{m/s})^2}{2(9.80\ \text{m/s}^2)} = \boxed{8.62\ \text{m}}$$



ANS. FIG. P6.17

(b) Let n be the force exerted by the rail.

Newton's second law gives

$$Mg + n = \frac{Mv^2}{r}$$

$$n = M\left(\frac{v^2}{r} - g\right) = M(2g - g) = \boxed{Mg,\ \text{downward}}$$

(c) $a_c = \dfrac{v^2}{r}$, or $a_c = \dfrac{(13.0\ \text{m/s})^2}{20.0\ \text{m}} = \boxed{8.45\ \text{m/s}^2}$

(d) If the force exerted by the rail is n_1,

then $n_1 + Mg = \dfrac{Mv^2}{r} = Ma_c$

$n_1 = M(a_c - g)$ which is < 0 since $a_c = 8.45\ \text{m/s}^2$